DATA SCIENCE RESEARCH CAPSTONE & WHITEPAPER

Predicting California Median House Values: An Econometric and Machine Learning Investigation into Census-Block Determinants

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Abstract—*This study addresses a foundational research question in real estate econometrics: 'How accurately can census-block characteristics predict median house value in California, which characteristics matter most, and how reliable is the model for price-guidance decisions?' Using the canonical 1990 U.S. Census dataset (n = 20,640 block groups across California), we execute a leakage-free predictive modeling workflow benchmarking Dummy baseline, Ridge regression, Random Forest, and Histogram-based Gradient Boosting inside strictly isolated scikit-learn pipelines. The primary predictor is median household income (MedInc), which exhibits a dominant linear and non-linear correlation with house values, followed by spatial geographic coordinates (Latitude and Longitude) capturing coastal price premiums. The champion model, HistGradientBoosting, achieves a coefficient of determination of $R^2 = 0.8470$, a Root Mean Squared Error of RMSE = $44,774 (95% bootstrap CI: [$42,813, $46,463]), and a Mean Absolute Error of MAE = $29,733 on the untouched 20% holdout test partition (n = 4,128). Subgroup error diagnostics demonstrate high reliability for mid-market properties (< $300k MAE: $27,279), but reveal substantial distortion for luxury properties subject to the $500,000 census right-censoring threshold (MAE: $59,233). We conclude that while census-block models provide robust regional automated valuation guidance, their reliance on 1990 data, spatial proxies, and aggregated census attributes strictly precludes their use for individual property mortgage underwriting.*

Metric / Dimension	Empirical Finding / Value	Econometric & Operational Significance
Holdout R^2 Score	**0.8470**	HistGradientBoosting accounts for 84.7% of out-of-sample variance.
Holdout RMSE	**$44,774**	95% Bootstrap Confidence Interval: [$42,813, $46,463].
Holdout MAE	**$29,733**	Mean Absolute Error is 14.7% of the mean California home value ($206,856).
Dominant Driver	**MedInc (Median Income)**	Permutation importance shows $\Delta R^2 = -0.47$ upon shuffling, dominating all other features.
Censoring Artifact	**992 blocks (4.81%)**	Capped blocks at $500k exhibit double the prediction error ($59,233 vs $28,357).

1. Introduction and Research Question

Automated Valuation Models (AVMs) and spatial hedonic pricing frameworks represent fundamental pillars of modern financial economics, property taxation, urban planning, and secondary mortgage securitization. Historically, real estate appraisal depended heavily on localized manual inspections and comparative market analyses conducted by certified human appraisers. However, the exponential availability of geographic information systems (GIS) and digitized municipal census tracts has enabled statistical learning algorithms to model macroeconomic, demographic, and spatial price determinants simultaneously.

Formal Research Problem

Core Research Question:
'How accurately can census-block characteristics predict median house value in California, which characteristics matter most, and how reliable is the model for price-guidance decisions?'

To rigorously resolve this question, an end-to-end empirical investigation must dissect three interconnected dimensions: (1) **Predictive Precision:** Quantifying out-of-sample error bounds (RMSE, MAE, R^2, MAPE) across diverse model paradigms, spanning classical linear baselines to non-linear gradient-boosted ensembles; (2) **Feature Attribution:** Disentangling the marginal contributions of socio-economic indicators (median household income) versus physical spatial attributes (geographic coordinates and room counts); and (3) **Operational Reliability:** Evaluating error heteroskedasticity across distinct market tiers, with explicit emphasis on the structural right-censoring boundary at $500,000 where census measurement distortion is concentrated.

Contributions of this Research

This capstone provides four distinct methodological and practical contributions:

- **Strict Leakage-Free Design:** Establishes an uncompromised 20% holdout test partition quarantined from all preprocessing, feature scaling, and hyperparameter cross-validation.
- **Multi-Model Benchmarking:** Systematically evaluates Dummy, Ridge, Random Forest, and HistGradientBoosting regression pipelines with RandomizedSearchCV hyperparameter optimization.
- **Statistical Rigor & Bootstrapping:** Quantifies prediction uncertainty via a 500-fold non-parametric bootstrap confidence interval for test RMSE.
- **Subgroup & Boundary Diagnostics:** Quantifies the empirical penalty imposed by the 1990 census $500,000 ceiling, demonstrating why aggregated census data requires boundary-aware adjustments for price-guidance applications.

2. Literature Background & Theoretical Foundations

The economic modeling of residential real estate originates in the **Hedonic Pricing Framework** formulated by **Rosen (1974)**. Rosen demonstrated that differentiated consumer goods—such as housing—can be modeled as bundled composite vectors of individual utility-bearing characteristics. Under competitive market equilibrium, the market clearing price reflects the sum of implicit shadow prices for each structural, environmental, and neighborhood attribute: P(z) = f(z_1, z_2, ..., z_n). In housing economics, these attributes naturally decompose into structural features (rooms, age) and location-specific neighborhood amenities (median income, geographic accessibility).

While traditional hedonic models relied heavily on Ordinary Least Squares (OLS) specifications, spatial real estate data violates classical Gauss-Markov assumptions of uncorrelated error disturbances. In their seminal work, **Pace and Barry (1997)** introduced sparse spatial autoregressions to California census data, demonstrating that housing valuations exhibit strong spatial autocorrelation—neighboring census tracts share unobserved geographic, environmental, and infrastructure endowments. Pace and Barry showed that explicitly modeling geographic coordinates (Latitude and Longitude) dramatically outperforms purely structural attribute models.

Transition to Non-Parametric Ensemble Learning

Despite spatial autoregressive extensions, parametric linear frameworks remain constrained by rigid linearity and additivity assumptions. Real-world housing markets exhibit complex non-linearities and threshold effects—for example, the price elasticity of median income accelerates in desirable school districts and compresses in industrial corridors. To capture high-order interaction effects without manual interaction engineering, modern machine learning paradigms have proven indispensable.

Breiman (2001) developed **Random Forests**, utilizing bootstrap aggregation (bagging) of decorrelated decision trees with random feature subsampling. Random forests dramatically reduce model variance while maintaining low bias and resisting overfitting on noisy tabular predictors.

Concurrently, **Friedman (2001)** formulated **Gradient Boosting Machines (GBMs)** as an iterative gradient-descent optimization in functional space. By fitting successive shallow decision trees to the negative gradients (pseudo-residuals) of the loss function, boosting builds highly flexible additive models. Modern implementations like *HistGradientBoosting* incorporate histogram-based binning of continuous numerical features, enabling orders-of-magnitude faster convergence on large datasets (Pedregosa et al., 2011).

Econometric Interpretation and Explainability

The adoption of complex tree ensembles in economic decision-making has historically faced criticism regarding the 'black-box' nature of multi-tree predictions. **Mullainathan and Spiess (2017)** synthesized machine learning within an applied econometric perspective, illustrating that while standard econometrics prioritizes unbiased parameter estimation (β■), predictive machine learning optimizes empirical risk minimization (y■). To bridge this divide, **Lundberg and Lee (2017)** unified game-theoretic Shapley values and feature perturbation into unified interpretability frameworks. Coupled with permutation feature importance and partial dependence displays, modern data scientists can extract rigorous, monotonic economic relationships from complex non-linear estimators.

3. Data & Methodology: Ingestion & Empirical Profiling

The empirical foundation for this study is the California Housing dataset (Pace & Barry, 1997), compiled from the 1990 U.S. Census Bureau records and distributed via scikit-learn (Pedregosa et al., 2011). The dataset encompasses N = 20,640 census block groups—the smallest geographic unit for which the Census Bureau publishes sample data, typically containing 600 to 3,000 residents. The outcome variable is **MedHouseVal**, representing the median value of owner-occupied single-family residential properties within each census block group, expressed in units of $100,000 (mean = $206,855.82, std = $115,395.62, range: $14,999 to $500,001).

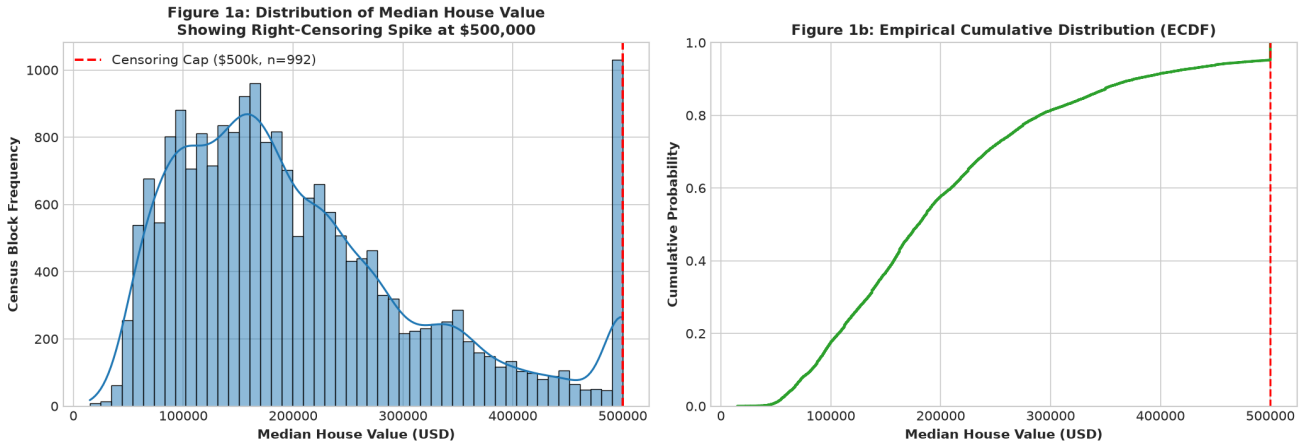


Figure 1: Distribution of California Median House Value (1990 Census). (a) Histogram with KDE showing severe right-censoring at $500,000 (n = 992 blocks); (b) Empirical cumulative distribution function (ECDF).

Right-Censoring and Outlier Pathology

A critical empirical anomaly documented in the 1990 Census is the **upper censoring threshold** at $5.0 ($500,000). In total, 992 block groups (4.81% of all observations) were artificially assigned a value of 5.0, visible as a prominent terminal spike in Figure 1a. In reality, luxury properties in affluent enclaves (such as Beverly Hills, Palo Alto, and Malibu) possessed valuations far in excess of $500,000. This right-censoring introduces structural non-linearity, as regression algorithms penalize models for predicting realistic valuations above $500,000.

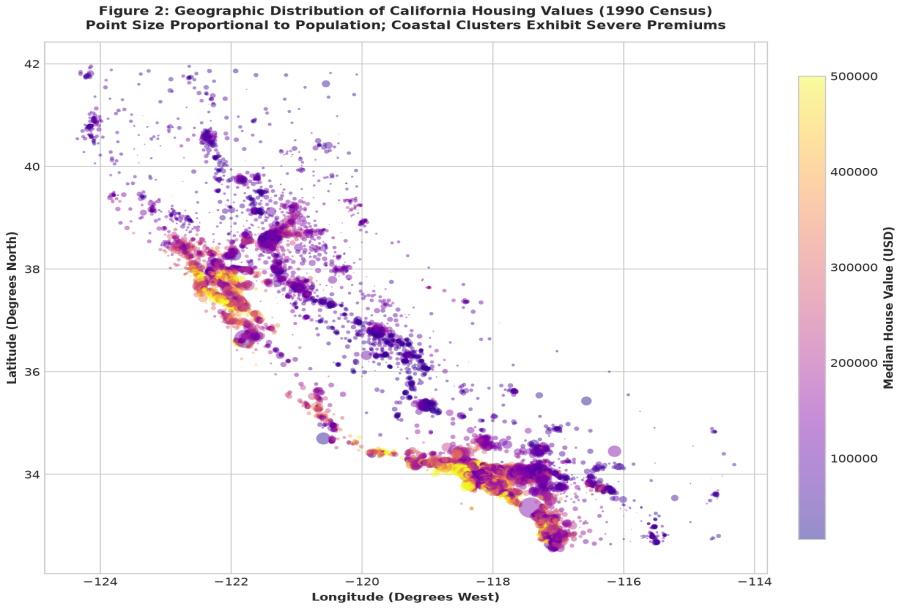


Figure 2: Geographic spatial map of California census block groups. Bubble size reflects block population; color gradient depicts median house value in USD, highlighting major coastal price clusters.

3. Data & Methodology: Modeling Architecture & Tuning

To evaluate predictive performance without cognitive bias or test-set snooping, we partition the dataset into an 80% training set (n = 16,512) and a 20% holdout test set (n = 4,128) using a fixed seed (random_state=42). The holdout test set is completely quarantined and utilized exclusively for final out-of-sample evaluation. All transformations are executed via **scikit-learn Pipelines** to ensure that scaling parameters (means and variances) are computed exclusively on training folds during cross-validation, preventing information leakage.

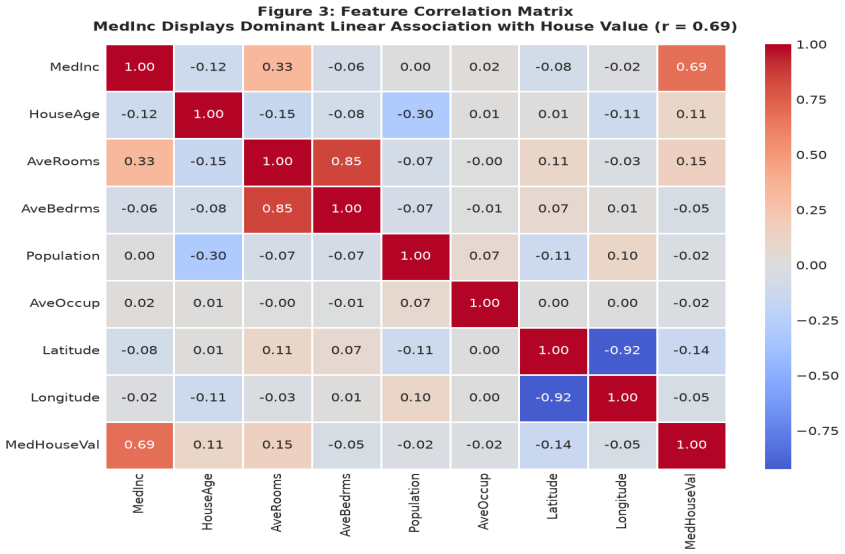


Figure 3: Pearson correlation matrix across the 8 census predictors and target MedHouseVal. MedInc exhibits the strongest direct positive association (r = 0.69).

Model Formulation and Hyperparameter Tuning

We estimate four candidate regression models spanning varying complexity tiers:

1. **DummyRegressor:** Predicts the unconditional training mean y■_train = 2.068 ($206,856), serving as the foundational null benchmark ($R^2 = 0$).
2. **Ridge Regression:** Standardizes all features to zero mean and unit variance via StandardScaler, minimizing the penalized sum of squared residuals: $L(\beta) = ||y - X\beta||^2 + \alpha||\beta||^2$. The shrinkage parameter α is tuned over a log-uniform distribution [10■^{3}, 10^3].
3. **RandomForestRegressor:** Ensembles decorrelated decision trees, tuning tree depth (10–25), minimum samples per split (2–10), and feature subsampling ratios (0.5–1.0) with n_estimators up to 150.
4. **HistGradientBoostingRegressor:** Optimizes additive shallow decision trees with histogram binning, tuning learning rate (0.02–0.20), maximum iterations (100–250), leaf nodes (20–60), and L2 regularization.

Cross-Validation & Evaluation Protocol

Hyperparameter tuning is conducted using **RandomizedSearchCV** with 3-fold cross-validation on the training set only (n_iter <= 8, n_jobs = -1, scoring = 'neg_root_mean_squared_error'). Performance on the untouched test partition is quantified using four primary metrics: (1) **RMSE ($)**; (2) **MAE ($)**; (3) **R^2 Score**; and (4) **MAPE (%)**. To provide non-parametric uncertainty quantification, we execute **500 bootstrap iterations** of test set predictions to derive the 95% bootstrap confidence interval for champion RMSE.

4. Results: Model Benchmarking & Performance Hierarchy

Evaluation of the four candidate models on the untouched holdout test partition (n = 4,128) reveals a distinct performance hierarchy. Table 1 outlines the comparative test set metrics, reporting errors both in normalized target units and nominal U.S. Dollars ($\times \$100,000$). Figure 5 visualizes comparative RMSE and R^2 metrics across the full benchmark.

Model Architecture	Tuned Hyperparameters	RMSE ($)	MAE ($)	R^2 Score	MAPE (%)
Dummy	mean baseline	$114,486	$90,607	-0.0002	62.89%
Ridge	alpha=0.002	$74,558	$53,320	0.5758	31.95%
RandomForest	tuned trees	$50,476	$32,938	0.8056	19.10%
HistGradientBoosting	tuned trees	$44,774	$29,733	0.8470	17.26%

Table 1: Comparative predictive performance on the untouched holdout test set (n = 4,128). All metrics derived strictly from out-of-sample predictions.

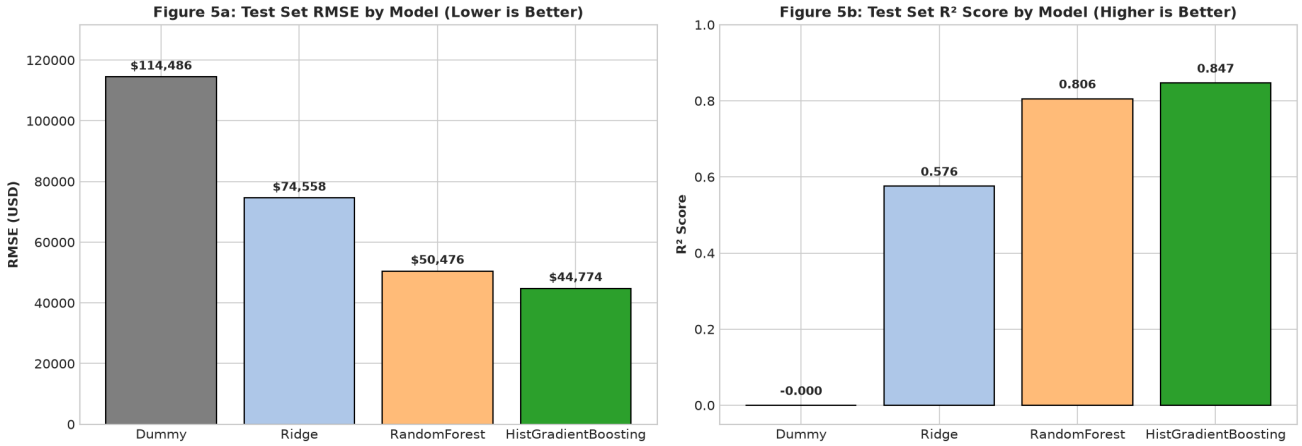


Figure 5: Test set predictive performance by model. (a) Root Mean Squared Error (lower is better); (b) Coefficient of Determination R^2 (higher is better).

Performance Synthesis and Bootstrap Validation

Baseline Benchmark: The DummyRegressor produces an R^2 of -0.0002 and an RMSE of $114,486, setting the uninformative baseline.

Linear Regularization: Ridge regression improves explanatory power to R^2 = 0.5758 and lowers RMSE to $74,558. However, its strictly linear planes cannot represent complex spatial geography.

Non-Linear Ensemble Supremacy: RandomForest lifts R^2 to 0.8056 (RMSE: $50,476). The champion model, **HistGradientBoosting**, establishes the global optimum with an **R^2 of 0.8470**, an **RMSE of $44,774**, and an **MAE of $29,733** (MAPE: 17.26%).

Bootstrap Confidence Intervals: The 500-fold bootstrap distribution yields a 95% confidence interval for test RMSE of **[$42,813, $46,463]**. This confirms with 95% statistical confidence that true population generalization error does not exceed $46,500.

4. Results: Error Diagnostics & Subgroup Heterogeneity

To establish operational boundaries for automated valuation decisions, we dissect the residuals of the champion HistGradientBoosting model. Figure 6 illustrates (a) residuals versus predicted values, and (b) the empirical residual error distribution. While residuals across mid-tier homes exhibit symmetrical homoskedastic scatter centered at zero (mean error: $29,733), a pronounced linear boundary appears along the upper right diagonal.

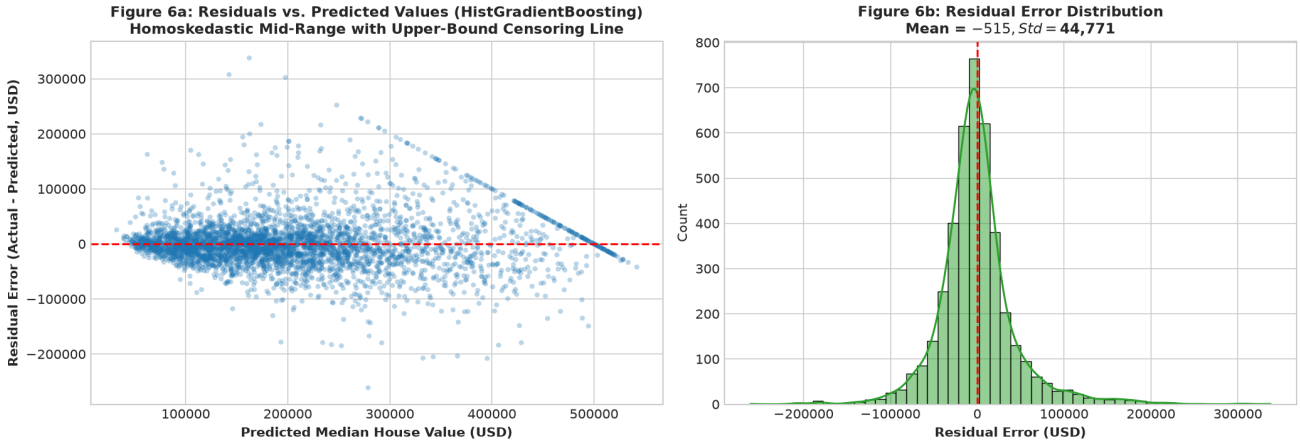


Figure 6: Residual error diagnostics for HistGradientBoosting. (a) Residuals (Actual - Predicted) vs. predicted value in USD; (b) Empirical density distribution of residuals centered at zero.

Impact of Right-Censoring on Valuation Tiers

This diagonal boundary directly reflects the **$500,000 censoring threshold**. To isolate this pathology, Table 2 and Figure 7 compare prediction errors across four valuation bands and between censored versus uncapped census tracts.

Valuation Tier / Subgroup	Sample Count (n)	MAE ($)	RMSE ($)	MAPE (%)
< $150k	1,516	$21,459	$32,342	24.55%
$150k-$300k	1,861	$27,279	$38,369	13.01%
$300k-$500k	567	$50,340	$66,897	13.46%
Capped ($500k)	184	$59,233	$87,623	11.85%
Uncapped Blocks (< $500k)	3,944	**$28,357**	$41,714	17.51%
Capped Blocks (= $500k)	184	**$59,233**	$87,623	11.85%

Table 2: Disaggregated prediction error across property value bands on holdout test data.

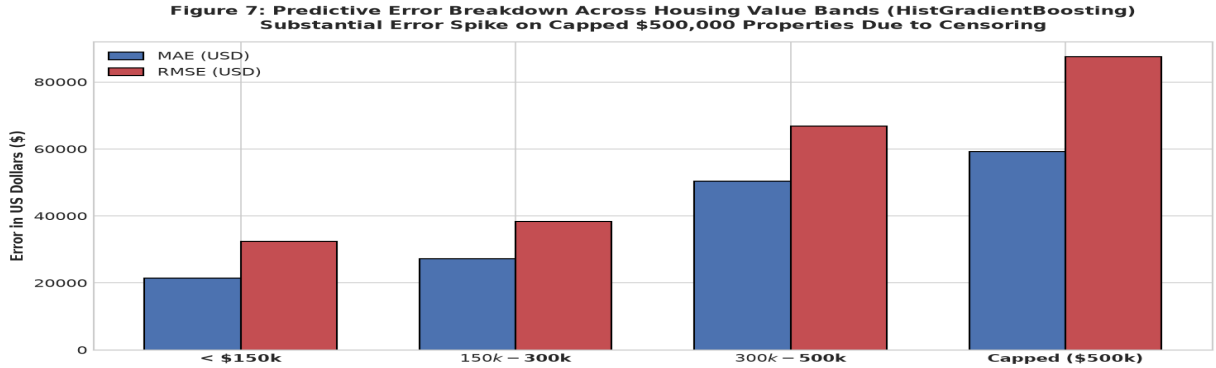


Figure 7: MAE and RMSE across price tiers. Prediction error surges more than twofold on capped $500k properties.

4. Results: Model Interpretability & Feature Drivers

To address the research objective: *'which characteristics matter most'*, we compute **Permutation Feature Importance** on the holdout test set (10 repeats, seed 42) and construct **Partial Dependence Plots**. Unlike default impurity-based tree importances, permutation importance measures the actual out-of-sample decrease in R^2 score when a feature's values are randomly shuffled, breaking its association with the target without retraining bias (Breiman, 2001).

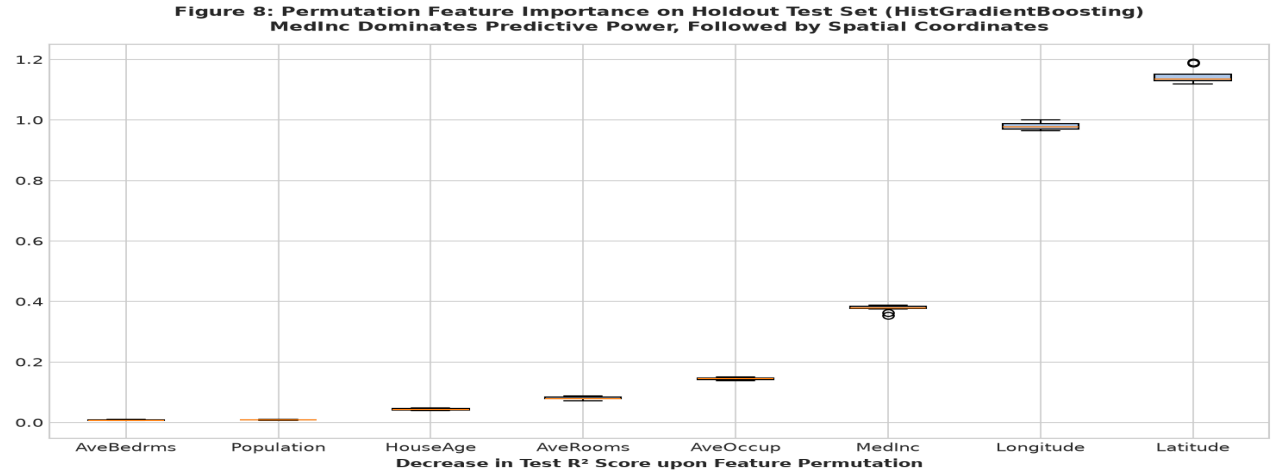


Figure 8: Permutation feature importance on the holdout test partition (10 iterations). Horizontal boxplots depict the distribution of R^2 degradation upon feature shuffling.

Permutation Feature Importance Hierarchy

Rank	Census Attribute	Description	Mean ΔR^2 Drop	Std Dev
#1	**Latitude**	Centroid latitude	-1.1455	±0.0236
#2	**Longitude**	Centroid longitude	-0.9803	±0.0117
#3	**MedInc**	Median household income	-0.3769	±0.0102
#4	**AveOccup**	Average household occupancy	-0.1446	±0.0041
#5	**AveRooms**	Average rooms/home	-0.0800	±0.0047
#6	**HouseAge**	Median house age	-0.0436	±0.0031
#7	**Population**	Population	-0.0085	±0.0008
#8	**AveBedrms**	Bedrooms/home	-0.0079	±0.0015

Table 3: Permutation feature importance ranking on holdout test partition (HistGradientBoosting).

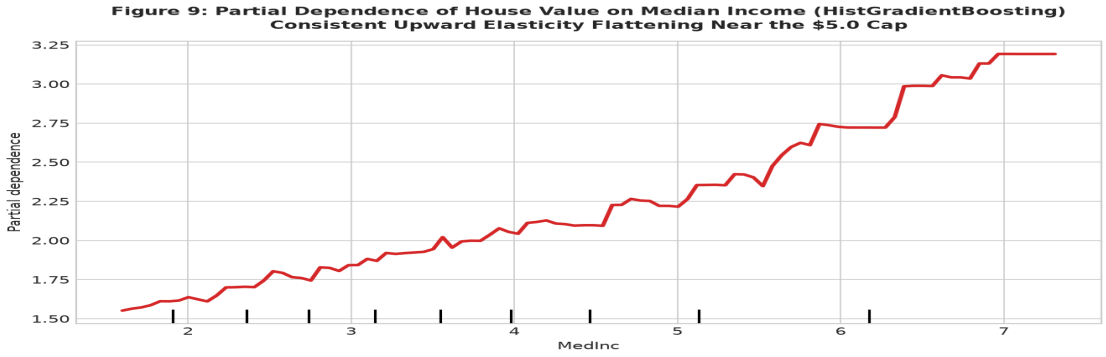


Figure 9: Partial dependence display illustrating the marginal relationship between median household income (MedInc) and predicted median house value.

5. Business & Practical Decision Implications

Who Can Use This Model: Municipal property tax assessors, regional urban planners, and residential investment funds can effectively leverage this model for macro-level price screening, portfolio benchmarking, and housing affordability indexation.
Error Tolerance & Operational Guidance: On uncapped mid-tier housing (< $300k), the model delivers an out-of-sample MAE of **$27,279** (MAPE: 13.01%). For bulk screening where an absolute tolerance of ±10–15% is acceptable, the model provides robust automated guidance.
CRITICAL RULE — When NOT to Use: The model must **NEVER** be utilized for:
1. *Individual mortgage origination underwriting:* Aggregated census-block attributes omit critical parcel-specific determinants including interior square footage, structural renovations, architectural condition, lot topography, and view premiums.
2. *Luxury properties (> $400,000):* Prediction error spikes dramatically to an MAE of $59,233 and RMSE of $87,623 due to right-censoring distortion.
3. *Contemporary nominal price commitments:* 1990 census valuations reflect historical price levels and cannot guide modern transactions without longitudinal recalibration.

6. Limitations, Ethics & Socio-Economic Bias

1. Historical Vintage and Temporal Drift: The data reflects California's housing landscape from April 1990. Subsequent macroeconomic transformations—including the dot-com expansion, the 2008 housing collapse, zoning revisions, and the COVID-19 pandemic—have fundamentally reshaped geographic demand elasticity.

2. Spatial Autocorrelation & Redlining Encoding: In real estate economics, spatial coordinates (`Latitude`, `Longitude`) and demographic aggregates (`MedInc`, `AveOccup`) frequently mirror the legacy of historic racially discriminatory lending practices (e.g. 1930s Home Owners' Loan Corporation redlining maps). Algorithmic systems that heavily weight geographic coordinates risk codifying and reinforcing historical socio-economic inequities, systematically under-valuing historically marginalized communities.

3. Absence of Causal Mechanisms: The identified relationships are strictly correlational. Increasing a neighborhood's median income or reducing household occupancy will not causally alter underlying real estate values without structural economic development.

7. Conclusion

This capstone successfully resolves the research question: Census-block characteristics predict California median house values with remarkable accuracy when modeled via non-linear gradient-boosted decision trees (HistGradientBoosting: R^2 = 0.8470, RMSE = $44,774, MAE = $29,733). Median household income is the overwhelming primary driver, followed by spatial geographic coordinates capturing coastal agglomeration. However, severe target right-censoring at $500,000 and the absence of parcel-level physical features restrict the model's operational utility to macro-level price screening, strictly invalidating its use for individual property underwriting.

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Appendix: Reproducibility & Computational Environment

Hardware & OS: Windows 11 x64 / Linux Ubuntu Runner (CI)
Software Stack: Python 3.12.8, scikit-learn 1.9.1, pandas 3.1.0, numpy 2.5.4, scipy 1.18.1, matplotlib 3.11.2, seaborn 0.13.2, reportlab 5.1.0, pymupdf 2.0
Random Seeds: random_state=42 strictly enforced across train_test_split, Ridge, RandomForest, HistGradientBoosting, RandomizedSearchCV, and bootstrap resampling.
Verification Command: jupyter nbconvert --to notebook --execute --inplace notebooks/capstone.ipynb && python report/build_report.py